



Stock prices of clean energy firms, oil and carbon markets: A vector autoregressive analysis[☆]

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ABSTRACT

Recent discussions of energy security and climate change have attracted significant attention to clean energy. We hypothesize that rising prices of conventional energy and/or placement of a price on carbon emissions would encourage investments in clean energy firms. The data from three clean energy indices show that oil prices and technology stock prices separately affect the stock prices of clean energy firms. However, the data fail to demonstrate a significant relationship between carbon prices and the stock prices of the firms.

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1. Introduction

Production costs of almost all clean (or alternative) energies are still high. Thus, these sources are rarely adopted in the market without subsidies. In the meantime, recent discussions of energy security and climate have attracted significant attention to clean energy. In this context, it is necessary to underscore the necessity of financial investments to support clean energy.

In the past 10 years, investments in alternative energy technologies have received attention. Moreover, public expenditures in the area of energy research, development and deployment (ERD&D) in Organisation for Economic Co-operation and Development (OECD)

countries showed a significant increase in the wake of the oil crises of the 1970s. These expenditures peaked in the early 1980s and declined thereafter (Gallagher et al., 2006).² In developed economies, it was the private sector and not the public sector that invested in ERD&D technology. Recent interest in climate change and resource scarcity caused another surge in alternate energy investments.

Equity and venture capital investments in alternative energy technologies are potentially promising sources for ERD&D support. Recent studies show that these new funding mechanisms are more effective than conventional research and development (R&D) resources in the sector. For example, Kortum and Lerner (2000) show that venture capital investment is 3–4 times more effective than R&D at stimulating patenting. Therefore, it is our desire to understand the financial mechanism behind clean energy.

One factor that stimulates such investment is oil prices. Oil prices increased from the January 2007 price of less than \$50, then declined to

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² Although the exact figures for private investment in ERD&D are not available, several observations support this position. The National Science Foundation's annual survey of industrial R&D indicates that (public and private) funds for industrial energy R&D showed an almost continual decline during the 1980s and 1990s, with 1999 levels that were about one-fifth of their peak values in 1980 in real terms. See http://www.nsf.gov/statistics/iris/research_hist.cfm?index=21 as quoted in Gallagher et al. (2006).

about one-third of the peak of \$147.30 in July 2008, and then increased again. The recent financial crisis has placed these clean energy issues in the background. Moreover, the stock prices of alternative energy sources might be falling relative to those of conventional energy sources, such as oil and gas, and relative to general economic stocks (e.g., DiFranza, 2010).

Rising oil prices increase the production costs of goods and services, dampen cash flow, and reduce stock prices.³ Previous studies found a positive association between rising oil prices and inflationary pressures on economies (e.g., Cunado and Perez, 2005; Darby, 1982; Fama, 1981). Rising oil prices and the resulting inflationary pressures affect the discount rate used in the equity pricing formula for valuing stock prices. Many studies find a negative and statistically significant relationship between oil price movements and stock prices (e.g., Cong et al., 2008; Henriques and Sadorsky, 2008; Huang and Masulis, 1996; Jones and GautamKaul, 1996; Park and Ratti, 2008; Sadorsky, 1999, 2001). However, there are asymmetry effects of oil prices on economic activities (e.g., Brown and Yucel, 2002; Hamilton, 1983, 1996; Jones and Leiby, 1996; Mork, 1989; Mork and Olsen, 1994). Although rising oil prices affect economic activities negatively, declining oil prices fail to stimulate economies.

This study investigates the relationship between oil prices and the prices of alternate energy stocks. The relationship between rising oil prices and the prices of alternate energy stocks is not clear, with the exception of the relationship depicted by Henriques and Sadorsky (2008), as discussed below. We hypothesize that the relationship between oil and alternate energy prices is positive, because rising oil prices encourage the substitution of alternate energy sources for conventional energy sources.

We also consider the relationship between technology stock prices and the prices of alternative energy products. This relationship exists because investors might see the stocks of alternate energy sources as similar to other technological stocks. Technological stocks are very sensitive to business cycles, and oil prices are an important component of business cycles. Therefore, the prices for technology stocks, clean energy stocks and general economic stocks should be positively associated when oil prices are increasing (see Henriques and Sadorsky, 2008).⁴ This hypothesis reflects the behavior of the stock market due to the business cycle effects of oil prices.

We expect concerns over global climate change to drive the growth of alternate energy sources, which are less carbon-intensive than conventional energy sources. To encourage the reduction of carbon emissions, the European Union Emission Trading System (EU-ETS) placed a price on carbon emissions under the cap and trade system. We hypothesize that higher carbon permit prices induce the development of alternate energy technologies and that there is a positive association between carbon permit prices and the stock prices of alternate energy sources. However, the recent economic downturn has reduced the prices of carbon permits. It is interesting to investigate the relationship between the carbon markets and the stocks of alternate energy sources.

There is little analysis of the relationship between clean energy firms, oil markets, and the carbon market with regard to financial performance (e.g., Managi and Okimoto, 2011). Henriques and Sadorsky (2008) study the relationship between clean energy stock prices and oil prices using the vector auto-regressive (VAR) approach over the period of January 3, 2001 to May 30, 2007. They find that shocks to oil prices have little significant impact on the stock prices of alternative energy companies. Managi and Okimoto (2011) further

extend their work econometrically using Markov-switching VAR. We intend to amplify the work of Henriques and Sadorsky (2008) by also considering carbon market data before and after oil price peaks using weekly data from the period of April 22, 2005, the date on which the carbon price data was first available, to November 26, 2008. This information is important because an investigation of alternative energies for a recent timeframe might identify a significant relationship between these energies and conventional energy sources. We develop and estimate a 5-variable lag-augmented VAR model containing data on the indices of stock prices of clean energy firms, technology stock prices, oil prices, carbon prices and the rate of interest. To account for the robustness of the results, we use various clean energy indices.⁵

The paper is organized as follows. Section 2 describes the methodology of the VAR model. Section 3 presents the data used in the study. Section 4 discusses the empirical results. We provide concluding remarks in Section 5.

2. Methodology

We consider the following vector auto-regression model of order p (or simply, VAR(p)):

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \varepsilon_t, \quad (1)$$

where y_t is a $(n \times 1)$ vector of endogenous variables, $c = (c_1, \dots, c_n)'$ is the $(n \times 1)$ intercept vector of the VAR, ϕ_i is the i th $(n \times n)$ matrix of autoregressive coefficients for $i = 1, 2, \dots, p$, and $\varepsilon_t = (\varepsilon_{1t}, \dots, \varepsilon_{nt})'$ is the $(n \times 1)$ generalization of a white noise process.

The main advantage of the VAR is that there is no need to specify which variables are the endogenous variables and which are the explanatory variables because in the VAR, all selected variables are treated as endogenous variables. That is, each variable depends on the lagged values of all selected variables and helps in capturing the complex dynamic properties of the data (Brooks, 2002).

Note that selection of appropriate lag length is crucial. If the chosen lag length is too large relative to the sample size, the degrees of freedom will be reduced and the standard errors of estimated coefficients will be large. If the chosen lag length is too small, then the selected lags in the VAR analysis may not be able to capture the dynamic properties of the data. The chosen lag length should be free of the problem of serial correlation in the residuals. Toda and Yamamoto (1995) propose a lag-augmented VAR (LA-VAR) testing procedure that is robust for the integration and co-integration properties of the data. LA-VAR avoids the pre-test bias. The usual lag-length selection criteria can be used to select the appropriate lag length (k) if the order of data integration does not exceed the true lag length of the model. That is, the VAR is estimated with $k + d_{max}$ lags, where d_{max} is the suspected maximum order of integration. K-coefficients are supposed to satisfy the linear or non-linear restrictions of standard asymptotic theory, and the last coefficient matrices are ignored because they are regarded as zeros.

To obtain robust results, it is necessary to determine the order of integration of each of the series and make them stationary. If each of the variables is found to be integrated on the order of 1 ($I(1)$), then a VAR model is estimated in the first differences of the series, and the conventional asymptotic theory is used for hypothesis testing. If the series is co-integrated rather than being used with the VAR, the error correction model (ECM) is used for estimation and hypothesis testing. It cannot be known a priori whether the series is stationary and not co-integrated. Pre-tests for unit roots and co-integration are performed before estimating a VAR model. ECM models are useful when all of the series of the same order are integrated. Yamada and Toda (1998), using Monte Carlo simulations, find that the LA-VAR

³ It is also important to note that increases in oil prices have a long-term effect on the economy. For example, a high growth rate in energy-saving technologies has been observed when oil prices are rising (e.g., Kumar and Managi, 2009).

⁴ Note that we do not consider indirect relationships between oil prices and prices for alternative energies; instead, we consider the relationship between oil prices and those for technology stocks. We then compare technology stock prices and the prices of alternative energy sources. We make this comparison because, as seen later in the data section, the simple correlation between technology stock prices and oil prices is so small that we can ignore its effect.

⁵ We apply three clean energy indices. These are the NEX, ECO, and SPGCE indices that we define in the next section.

model has better size stability relative to the ECM model, although the former approach has lower predictive power in comparison to the latter approach.

In a VAR model, a large number of coefficients are estimated, and often the estimated coefficients appear to be lacking in statistical significance. Thus, it is not interesting to examine the coefficients themselves. The VAR models are used to test Granger causality, i.e., whether the lags of one variable explain the current value of some other variable, and to describe the dynamics of the data (Brooks, 2002). Impulse response functions and forecasting variance decomposition analysis are used to explain the dynamic effects of the shocks on endogenous variables. Impulse response functions are used to test the impact of an innovation in one variable on the current and future values of another variable. If the error terms in the VAR are serial-correlated, the impulse responses are also correlated, and this effect makes it difficult to interpret the impulse responses. Orthogonalized impulse-response functions, using the Cholesky decomposition, are used to solve this problem. Orthogonalized impulse-response functions require an ordering of the variables in the system, as this method of orthogonalization involves the assignment of contemporaneous correlation only to specific series. Thus, the first variable in the ordering is not contemporaneously affected by shocks to the remaining variables, but shocks to the first variable do affect the other variables in the system. The second variable contemporaneously affects the other variables (with the exception of the first one), but it is not contemporaneously affected by them, and so on. Finding an appropriate ordering of variables is difficult, and the resulting impulse responses may not be sufficiently robust for the ordering of the variables in the VAR. Pesaran and Shin (1998) propose the use of a generalized impulse response that is not dependent on the ordering of variables in the VAR. We use generalized impulse responses to investigate the impact of oil and carbon price changes on the stock prices of clean energy firms. Variance decomposition analysis forecasts the variances caused by various variable shocks and is used to estimate the importance of various structural shocks.

Our basic VAR model has five stationary variables: the log of clean energy companies' stock prices, high technology firms' stock prices, oil prices, carbon prices and interest rates.

3. Data

In this paper, we use a weekly five-variable VAR to study the relationships between clean energy stock prices, the stock prices of technology companies, oil prices and carbon prices. The variables considered for the model are the following: the stock index of clean

energy, expressed as NEX, ECO and SPGCE, the index of the prices of technology stocks (PSE), oil prices, prices of carbon allowances traded at the European Emission Trading and short-term interest rates. The carbon price is not integral to global markets. However, it affects the prices of the Clean Development Mechanism (CDM), and therefore, we expect that the carbon price is a proxy for global prices. In addition, two of the three clean energy indices are incorporated into the global index, while one is US-oriented. The US did not ratify the Kyoto protocol and thus is not directly linked to the CDM. We expect, however, that many of the firms in the index are considered to be in the global market and will be affected by the carbon price. The weakness of this data limitation is discussed later. The stock market performance of clean energy firms is measured through the following three indices with data obtained from Bloomberg.

3.1. Global 100

3.1.1. NEX

The Wilder Hill New Energy Global Innovation Index is a modified dollar-weighted index of publicly traded companies active in renewable and low-carbon energy. This index benefits from responses to climate change and energy security concerns. Most of the index members quoted are from outside the US. The benchmark value of 100 is from data as of December 30, 2002. This index is created by Wilder Hill New Energy Finance, LLC: DJI.

3.2. US 100

3.2.1. ECO

The Wilder Hill Clean Energy Index is a modified equal-dollar-weighted index composed of publicly traded companies whose businesses stand to benefit substantially from societal transition toward the use of cleaner energy and conservation. The index benchmark value was 100 at the close of trading on December 30, 2002. This index is disseminated by the American Stock Exchange (AMEX).

3.3. Global 30

3.3.1. SPGCE

The S&P Global Clean Energy Index provides liquid and tradable exposure to 30 companies from around the world that are involved in clean energy-related businesses. This index is a modified capitalization-weighted index comprised of a diversified mix of clean energy production and clean energy equipment and technology companies.

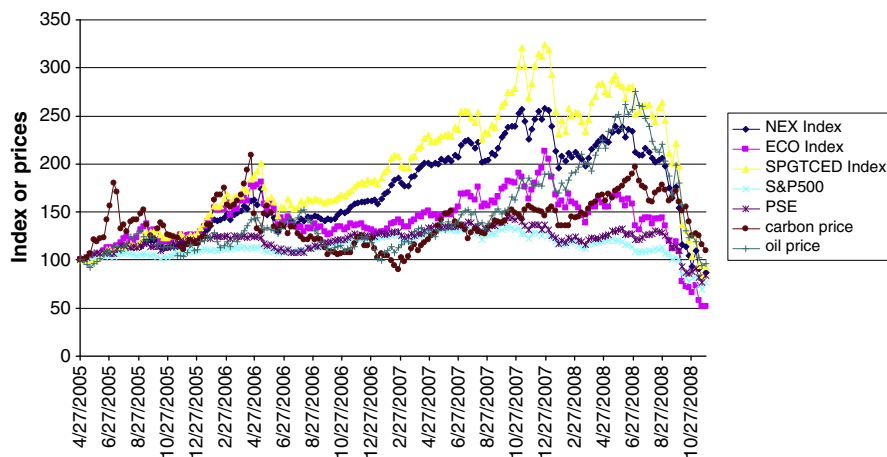


Fig. 1. Stock prices (alternative energy, technology, and general), carbon price and oil price.

Table 1
Correlation matrix.

	NEX	ECO	SPGCE	Carbon price	Oil price	PSE	Interest rate	S&P 500
NEX	1							
ECO	0.966	1						
SPGCE	0.912	0.943	1					
Carbon price	0.410	0.232	0.134	1				
Oil price	0.587	0.424	0.312	0.890	1			
PSE	0.879	0.922	0.867	0.073	0.221	1		
Interest rate	0.581	0.712	0.718	−0.339	−0.214	0.816	1	
S&P 500	0.812	0.895	0.857	−0.100	0.056	0.964	0.895	1

Table 2
Descriptive statistics of weekly returns.

Variable	Obs	Mean	Std. Dev	Min	Max	Skewness	Kurtosis	Sharpe ratio
NEX index	187	0.03	4.54	−25.15	18.21	−1.37	9.76	−0.01
ECO index	187	−0.21	5.17	−29.05	11.67	−1.47	8.73	−0.05
S&P GTCED index	187	0.13	5.63	−30.88	23.94	−1.46	11.22	0.01
Carbon price	187	0.23	5.84	−28.98	18.34	−0.69	7.02	0.005
Oil price	187	0.10	4.97	−17.22	18.59	−0.15	3.88	0.03
PSE index	187	−0.06	2.59	−14.98	8.46	−1.42	9.38	−0.05
S&P 500	187	−0.11	2.40	−15.17	10.05	−1.65	13.41	−0.08

Descriptive statistics are presented for continuously compounded weekly returns (April 22, 2005 to November 26, 2008).

The behavior of stock market prices in the 1990s shows that investors may consider investments in clean energy firms as being similar to those of other high technology firms (Henriques and Sadosky, 2008). This comparison requires data on the stock market performance of high technology firms for investigation of the relationship between their stock prices and those of clean energy firms. We use the Pacific Stock Exchange (PSE) technology index to measure the stock market performance of high technology firms.

3.3.2. PSE

The Arca Tech 100 index, with a pure focus on technology, includes firms listed on leading stock exchanges and on over-the-counter facilities. It is a price-weighted, broad-based index comprising 100 listed and over-the-counter stocks from 15 different industries, including computer hardware, software, semiconductors, telecommunications, data storage and processing, electronics and biotechnology (see <http://www.nyse.com/marketinfo/indexes/pse.shtml>).

Table 3
Estimation of market risks from the multifactor model.

	NEX	ECO	SPGCE	PSE
Carbon price	−0.356 (−4.04) ***	−0.055 (−1.11)	0.199 (3.53) ***	0.333 (6.94) ***
Oil price	1.07 (19.19) ***	0.527 (13.25) ***	0.484 (13.47) ***	0.241 (6.24) ***
Rate of interest	0.219 (10.75) ***	−0.138 (−6.87) ***	0.330 (25.28) ***	0.171 (8.79) ***
S&P 500		2.11 (21.18) ***	0.939 (9.74) ***	0.227 (17.14) ***
Constant	1.90 (8.66) ***	−11.392 (−17.83) ***	2.142 (15.18)	3.192 (12.83) ***
Adj R ²	0.745	0.926	0.884	0.762
Observations	188	188	188	188

Values in parentheses contain heteroskedasticity and autocorrelation robust t-statistic.

***, **, * denote a test statistics is statistically significant at the 1%, 5%, or 10% level of significance.

3.3.3. Oil price

Oil, a conventional fossil fuel energy, is the most widely traded physical commodity in the world. However, environmental and energy security concerns require the substitution of oil by clean energy. Furthermore, rising oil prices should encourage investments in clean energy firms. In this paper, we measure oil prices using the average of the closing prices of the nearest contract on the West Texas Intermediate (WTI) and on the BRENT crude oil future contract (see http://tonto.eia.doe.gov/dnav/pet/xls/PET_PRI_SPT_S1_D.xls).

3.3.4. Carbon price

We hypothesize that setting a price on carbon would help in stabilizing carbon emission concentrations in the atmosphere. To investigate whether carbon prices have encouraged investments in clean energy firms, we use carbon allowance price future contracts ending on December 2012 with data from the European Emission Trading (see <http://www.ecx.eu/ECX-Historical-Data>). The EU-ETS lists several different carbon future prices because there are different vintages of carbon being traded at any one time. Daskalakis et al. (2009) find that the carbon spot price is likely to be characterized by jumps and non-stationarity. However, Trück et al. (2007) showed that futures that are written within the first phase and expire within the second have significant convenience yields. Therefore, we used future prices for the contracts ending in December 2012.

3.3.5. Interest rate

Previous research shows a significant relationship between stock price movements and interest rates (e.g., Sadosky, 1999, 2001). We use the yield on a 3-month US Treasury bill as a variable to investigate the relationship between the stock prices of clean energy firms and interest rates. Interest rate data are obtained from the Federal Reserve Board of St. Louis (see <http://research.stlouisfed.org/fred2/>).

The sample period is from April 22, 2005, the date from which the data on carbon price are available, to November 26, 2008, for a total of 188 available weekly observations. For the stock market prices and the prices of oil and carbon, we use the data from the Wednesday closing. Wednesday closing prices are used because, in general, there are fewer holidays on Wednesdays relative to Fridays. Any missing data from a Wednesday closing are replaced with closing prices from the most recent trading session.

Fig. 1 shows a time series plot of the NEX, ECO, SPGCE, PSE, S&P 500, oil price and European Union CO₂ allowance prices. For the sake of comparison, each series is set equal to 100 on April 22, 2005. We observe similar patterns in the stock prices of clean energy firms (NEX, ECO, and SPGCE), high technology firms (PSE) and general stocks (S&P 500). All of the stock price indices are highly correlated (see Table 1 for the correlation matrix). The prices of clean energy rose at a higher rate and were more volatile than general stock prices or

Table 4
Unit root test.

	Levels			First differences		
	ADF test	DF-GLS test	PP test	ADF test	DF-GLS test	PP test
ln(NEX)	1.923	−0.435	2.237	−13.498***	−3.489***	−13.551***
ln(ECO)	1.474	−0.277	1.733	−12.884***	−4.157***	−12.882***
ln(SPGCE)	1.475	−0.563	1.945	−13.577***	−2.839**	−13.589***
ln(carbon price)	−2.431	−1.939	−2.562	−11.46***	−5.223***	−11.285***
ln(oil price)	0.207	−2.002	−0.342	−12.755***	−3.344**	−12.881***
ln(PSE)	0.127	−1.303	−0.155	−12.282***	−3.166**	−12.341***
ln(rate of interest)	−1.047	−0.21	−0.652	−14.6***	−3.118**	−14.862***
ln(S&P 500)	0.724	−0.816	1.071	−13.668***	−2.543	−13.68***

Unit roots are tested using Augmented Dickey and Fuller (ADF), DF-GLS and Phillips and Perron (PP) unit root tests. All unit root tests regressions include an intercept and time trend variable. ***, **, * denote a test statistics is statistically significant at the 1%, 5%, or 10% level of significance.

Table 5
Lag selection-order criteria.

Lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
<i>NEX</i>								
0	398.7				9.10E-09	−4.33	−4.29	−4.24
1	1480.0	2162.6	25	0	8.30E-14	−15.93	−15.72	−15.406*
2	1504.6	49.1	25	0.003	8.30E-14	−15.93	−15.54	−14.96
3	1559.7	110.2	25	0	6.00E-14	−16.26	−15.69	−14.85
4	1610.8	102.3	25	0	4.50E-14	−16.55	−15.798*	−14.70
5	1648.6	75.55*	25	0	3.9e-14*	−16.6877*	−15.76	−14.40
6	1658.7	20.2	25	0.735	4.70E-14	−16.52	−15.42	−13.80
<i>ECO</i>								
0	446.9				5.40E-09	−4.86	−4.82	−4.77
1	1477.7	2061.5	25	0	8.50E-14	−15.91	−15.6947*	−15.3807*
2	1497.2	38.9	25	0.037	9.00E-14	−15.85	−15.46	−14.88
3	1537.6	80.9	25	0	7.60E-14	−16.02	−15.45	−14.61
4	1576.7	78.1	25	0	6.60E-14	−16.17	−15.42	−14.32
5	1605.9	58.558*	25	0	6.3e-14*	−16.2189*	−15.29	−13.93
6	1623.8	35.7	25	0.077	6.80E-14	−16.14	−15.03	−13.41
<i>SPGCE</i>								
0	366.8				1.30E-08	−3.98	−3.94	−3.89
1	1440.5	2147.3	25	0	1.30E-13	−15.50	−15.2853*	−14.9713*
2	1464.5	48.1	25	0.004	1.30E-13	−15.49	−15.10	−14.52
3	1509.0	89.0	25	0	1.00E-13	−15.70	−15.13	−14.29
4	1559.8	101.7	25	0	7.90E-14	−15.99	−15.24	−14.14
5	1600.2	80.757*	25	0	6.7e-14*	−16.156*	−15.23	−13.87
6	1611.0	21.7	25	0.653	7.90E-14	−16.00	−14.89	−13.27

stock prices of high technology firms. During the financial crisis, the fall in the prices of clean energy stocks was greater than the fall in the prices of general stocks or in the prices of high technology firms' stocks. The correlation between the performance of clean energy firms (NEX, ECO and SPGCE) and oil price is 0.59, 0.42 and 0.31, respectively. The correlation between the performance of the clean energy firms and carbon price is 0.41, 0.23, and 0.13, respectively.

Table 2 shows the average weekly continuously compounded returns on investments in clean energy firms, high technology firms, general stocks, carbon permits and oil futures. The average annual returns are obtained by multiplying the weekly returns by a factor of 52. The average annualized compounded returns on investments in clean energy firms were 1.73%, −10.82% and 6.71% for the NEX, ECO and SPGCE, respectively. The average annualized returns on general stocks and on high technology firms' stocks were −5.81% and −3.22%, respectively. At the same time, the average annualized returns on investments in oil and carbon futures were 5.39% and 11.87%, respectively, and the monthly return on the US Treasury bill was 3.97%. These average annualized returns reflect the results of investing in strong carbon markets and in the stocks of clean energy firms using the SPGCE index and oil market data over this time period. We use the Sharpe ratio to measure the risk-adjusted returns. A higher value in the Sharpe ratio indicates a lower risk and higher

Table 6
Vector auto-regressive model fit.

Equation	Parameters	RMSE	R ²	chi ²	P>chi ²
<i>NEX</i>					
lnex	31	0.038	0.984	10962.65	0
lpse	31	0.023	0.957	4033.00	0
lbrent	31	0.047	0.972	6271.48	0
lcarbon	31	0.058	0.897	1589.09	0
lrate	31	0.107	0.968	5446.05	0
<i>ECO</i>					
leco	31	0.047	0.959	4297.267	0
lpse	31	0.023	0.957	4047.599	0
lbrent	31	0.046	0.973	6671.961	0
lcarbon	31	0.056	0.902	1665.382	0
lrate	31	0.111	0.966	5129.855	0
<i>SPGCE</i>					
lspg	31	0.048	0.981	9575.491	0
lpse	31	0.024	0.956	3964.034	0
lbrent	31	0.046	0.973	6489.552	0
lcarbon	31	0.057	0.898	1609.607	0
lrate	31	0.107	0.968	5489.153	0

lnex: log of NEX, leco: log of ECO, lspg: log of SPGCE, lbrent: log of oil price, lcarbon: log of carbon price, lpse: log of PSE, and lrate: log of interest rate.

Table 7
Lagrange-multiplier test.

lag	dlnex		dleco		dlspg	
	chi ²	Prob>chi ²	chi ²	Prob>chi ²	chi ²	Prob>chi ²
1	22.73	0.59	26.95	0.36	33.84	0.11
2	25.58	0.43	29.58	0.24	33.11	0.13
3	38.32	0.04	37.61	0.05	44.67	0.01
4	18.99	0.80	26.98	0.36	15.28	0.93
5	26.12	0.40	15.38	0.93	22.49	0.61
6	34.37	0.10	36.35	0.08	38.75	0.04

H₀: no autocorrelation at lag order.

lnex: log of NEX, leco: log of ECO, and lspg: log of SPGCE.

returns. Table 2 shows that investing in the oil futures market followed by the SPGCE and carbon markets brought better returns than was the case for the other above-mentioned portfolios.

4. Empirical results

To investigate the market risk of investing in the stocks of clean energy firms, we use a multifactor model following Sadosky (2001). In a multifactor model, dependent variables are stock returns for clean energy firms or high technology firms. Independent variables are stock market returns measured by the S&P 500, risk factors for the oil and carbon price markets, and interest rate changes. Table 3 presents the results of multifactor models using ordinary least squares (OLS). The estimated coefficients on the market returns variable indicate that the NEX and SPGCE indices are approximately twice as risky as the S&P 500 index. However, the ECO and PSE indices are as risky as investments in the broad-based market (S&P 500). The interest rate variable is a significant risk factor for all of the indices. Oil price returns are a significant risk factor for clean energy stocks but not for the stocks of high technology firms. Carbon price returns are an equally significant variable for all of the stocks of clean energy firms and high technology firms. Note that multifactor models are useful for

investigating these relationships when all of the variables are measured contemporaneously and dynamic relationships are captured and explored using VAR models.

All of the variables are expressed in natural logarithms to reduce unwanted variability (heteroskedasticity) in the data. Before analyzing the effects of oil and carbon prices on the stock prices of clean energy firms, we proceed to investigate the stochastic properties of the series considered in the model. We do so by analyzing their order of integration on the basis of a series of unit root tests. Specifically, we perform the Augmented Dickey–Fuller (ADF), DF-GLS and Phillips–Perron (PP) tests. The results of these formal tests are summarized in Table 4, indicating that the first differences of all five variables are stationary.

To select a suitable lag length, different tests are considered: the modified Likelihood Ratio test (Sims, 1980), as well as the Akaike, Schwarz and Hannan–Quinn tests. When there is conflict between different tests, the optimal lag length is chosen using the Likelihood Ratio test. The chosen lag length (k) is five (Table 5). The VAR is estimated, following Toda and Yamamoto (1995), using six lags, as all the data series are integrated in order 1.

The VAR model fit values are given in Table 6. The model fit values show that the VAR fits well in all of the equations and that all of the fits are statistically significant. The R² values show that the explanatory variables are able to significantly explain the variations in the dependent variables. The root of the mean square error (RMSE) value is low, indicating a tighter-fitting model. The Lagrange Multiplier (LM) test statistics, given in Table 7, show the absence of serial correlation. Overall, the results show that the model fits the data well and that it can be used for hypothesis testing and analysis of the dynamic properties of the data. We also find that all of the eigenvalues lie inside the unit circle and that all three VAR models satisfy the stability condition.

Granger causality Wald test statistics, using the five variable LA-VAR model, show that the variation in all three indices of clean energy stocks is explained by past movements in oil prices, stock prices of high technology firms and interest rates (Table 8). Note that in all three models of clean energy stocks, we could not reject the null

Table 8
Granger causality Wald tests.

Equation	Excluded	lnex		leco		lspgce	
		chi ²	Prob>chi ²	chi ²	Prob>chi ²	chi ²	Prob>chi ²
lnex/leco/lspg	lpse	26.48	0.00	17.05	0.01	21.56	0.00
lnex/leco/lspg	lbrent	20.57	0.00	14.31	0.03	13.87	0.03
lnex/leco/lspg	lcarbon	7.06	0.32	6.60	0.36	6.02	0.42
lnex/leco/lspg	lrate	37.60	0.00	34.74	0.00	51.02	0.00
lnex/leco/lspg	ALL	121.01	0.00	101.02	0.00	128.14	0.00
lpse	lnex/leco/lspg	11.58	0.07	12.25	0.06	8.41	0.21
lpse	lbrent	15.81	0.02	16.28	0.01	13.54	0.04
lpse	lcarbon	4.38	0.63	5.72	0.46	5.46	0.49
lpse	lrate	24.60	0.00	22.83	0.00	21.72	0.00
lpse	ALL	82.08	0.00	82.99	0.00	77.76	0.00
lbrent	lnex/leco/lspg	16.79	0.01	29.13	0.00	23.51	0.00
lbrent	lpse	9.72	0.14	8.67	0.19	11.31	0.08
lbrent	lcarbon	6.87	0.33	7.06	0.32	7.07	0.31
lbrent	lrate	15.68	0.02	28.14	0.00	18.04	0.01
lbrent	ALL	45.42	0.01	59.53	0.00	53.11	0.00
lcarbon	lnex/leco/lspg	1.47	0.96	9.38	0.15	3.60	0.73
lcarbon	lpse	4.79	0.57	4.04	0.67	8.71	0.19
lcarbon	lbrent	6.09	0.41	8.40	0.21	7.63	0.27
lcarbon	lrate	4.68	0.59	12.82	0.05	5.91	0.43
lcarbon	ALL	33.71	0.09	43.00	0.01	36.20	0.05
lrate	lnex/leco/lspg	21.85	0.00	10.39	0.11	23.41	0.00
lrate	lpse	23.45	0.00	23.04	0.00	24.01	0.00
lrate	lbrent	13.06	0.04	12.54	0.05	10.43	0.11
lrate	lcarbon	4.57	0.60	4.39	0.62	5.07	0.53
lrate	ALL	117.54	0.00	100.71	0.00	119.83	0.00

lnex: log of NEX, leco: log of ECO, lspg: log of SPGCE, lbrent: log of oil price, lcarbon: log of carbon price, lpse: log of PSE, and lrate: log of interest rate.

hypothesis that carbon prices and the stock prices of clean energy companies are not related from a Granger causality perspective. This circumstance might be due to two reasons: (1) we consider the carbon price to be traded in the European market, whereas the clean energy stocks under study are traded at the United States stock exchanges. Moreover, US firms using conventional carbon-intensive fuels are not under regulation, and/or (2) the prevailing future carbon prices are low and are not able to internalize the carbon externalities. Therefore, they are not able to create a stimulus for the switch from

conventional fossil fuels to clean low-carbon technologies. The Wald tests statistics also indicate that the variation in the indices of clean energy stocks cannot be explained by any single independent variable.

Technology stock prices are influenced by the clean energy stocks represented by the NEX and ECO indices and by the variation in oil prices and interest rates. The Wald tests statistics reveal that carbon price is not able to influence the prices of technology stocks. Note that the interest rate Granger affects the stock prices of high technology

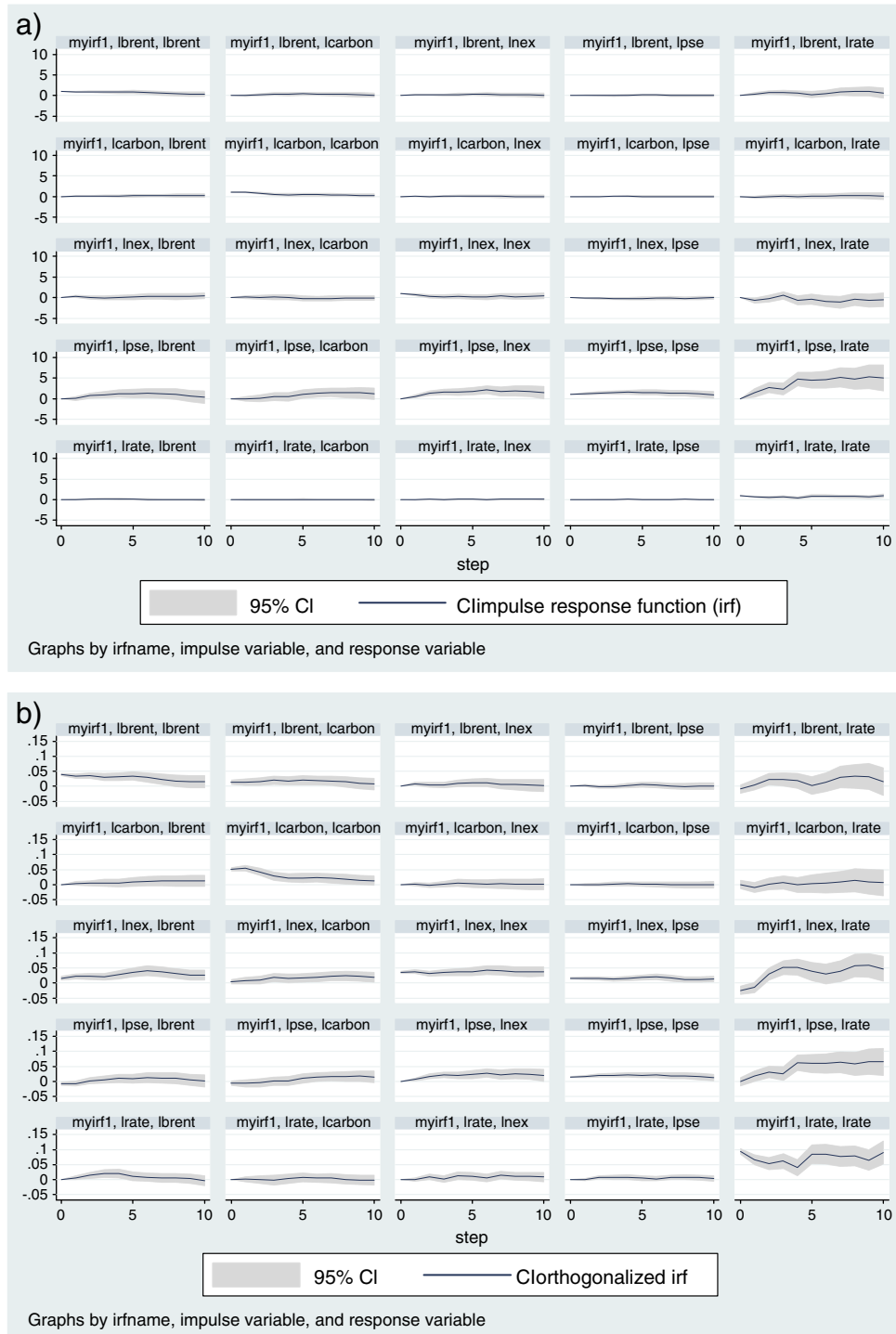


Fig. 2. a: impulse response functions (NEX). lnex: log of NEX, leco: log of ECO, lspg: log of SPGCE, lbrent: log of oil price, lcarbon: log of carbon price, lpse: log of PSE, and lrate: log of interest rate. b: orthogonal impulse response functions (NEX). lnex: log of NEX, leco: log of ECO, lspg: log of SPGCE, lbrent: log of oil price, lcarbon: log of carbon price, lpse: log of PSE, and lrate: log of interest rate.

firms and clean energy firms. This finding is consistent with the existing literature in the field. The identical relationships between the high technology firms' stock prices and oil price and between the clean energy firms' stock prices and oil price support the hypothesis that investors view the stocks of clean energy sources in the same way that they view the stocks of high technology firms. The relationship between the PSE and the different indices of clean firms also indicates that the relationship depends on the constituents of the clean energy stock index.

The Granger causality tests show that oil prices are influenced by the lagged values of interest rates. However, carbon prices are not influenced by any of the variables considered in the present study. Contrary to the findings of [Henriques and Sadorsky \(2008\)](#), we find that the stocks of high technology firms and clean energy firms influence the movements in interest rates. Interest rates are also influenced by the oil price. That is, the movements in interest rates are influenced not only by economic growth and inflation in the economy but also by the movements in markets.

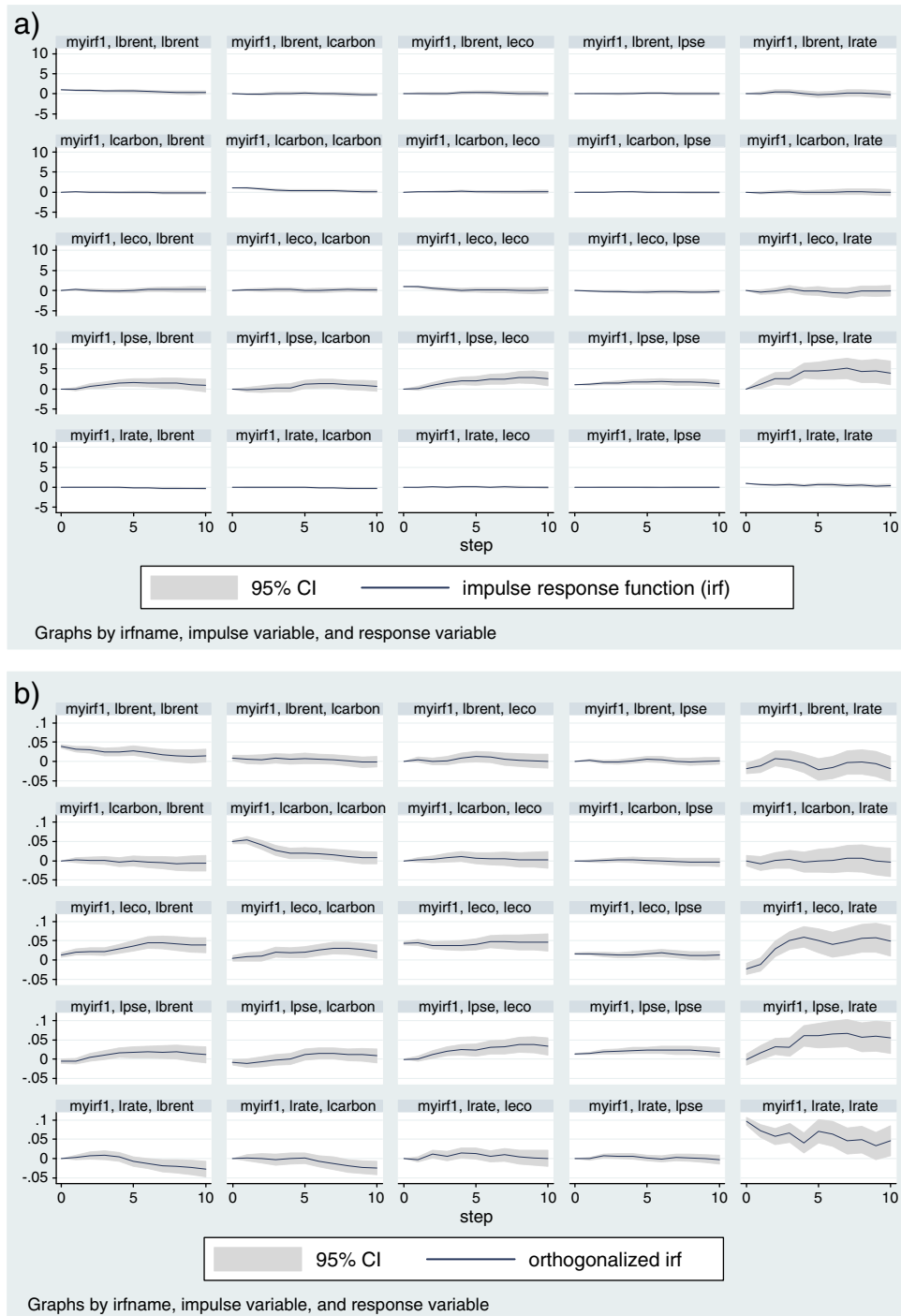


Fig. 3. a: impulse response functions (ECO). Inex: log of NEX, leco: log of ECO, lspg: log of SPGCE, lbrent: log of oil price, lcarbon: log of carbon price, lpse: log of PSE, and lrate: log of interest rate. b: orthogonal impulse response functions (ECO). Inex: log of NEX, leco: log of ECO, lspg: log of SPGCE, lbrent: log of oil price, lcarbon: log of carbon price, lpse: log of PSE, and lrate: log of interest rate.

We examine the effects of oil and carbon price shocks, coupled with the shocks to the stock prices of high technology firms and to interest rates, on the stocks of clean energy indices. We apply both the generalized impulse response function and the orthogonalized impulse response function. The impulse response function is a dynamic function composed of the partial derivatives of clean energy stocks at a given time. It goes hand in hand with the explanatory variable shock at each period in the past, possibly beginning with the contemporaneous period. The sum of the impulse response coeffi-

cients for a shock at a specific time yields the equivalent of a cumulative explanatory variable: clean energy stock price elasticity for a single period shock. Figs. 2–4 present the generalized impulse response functions and the orthogonalized impulse response functions of changes in clean energy stock prices to one standard deviation shock in the other VAR variables.

Initially, the stock prices of clean energy firms respond positively to a shock to their own prices, and this response continues through 2 weeks into the future. The most dramatic change is the impulse

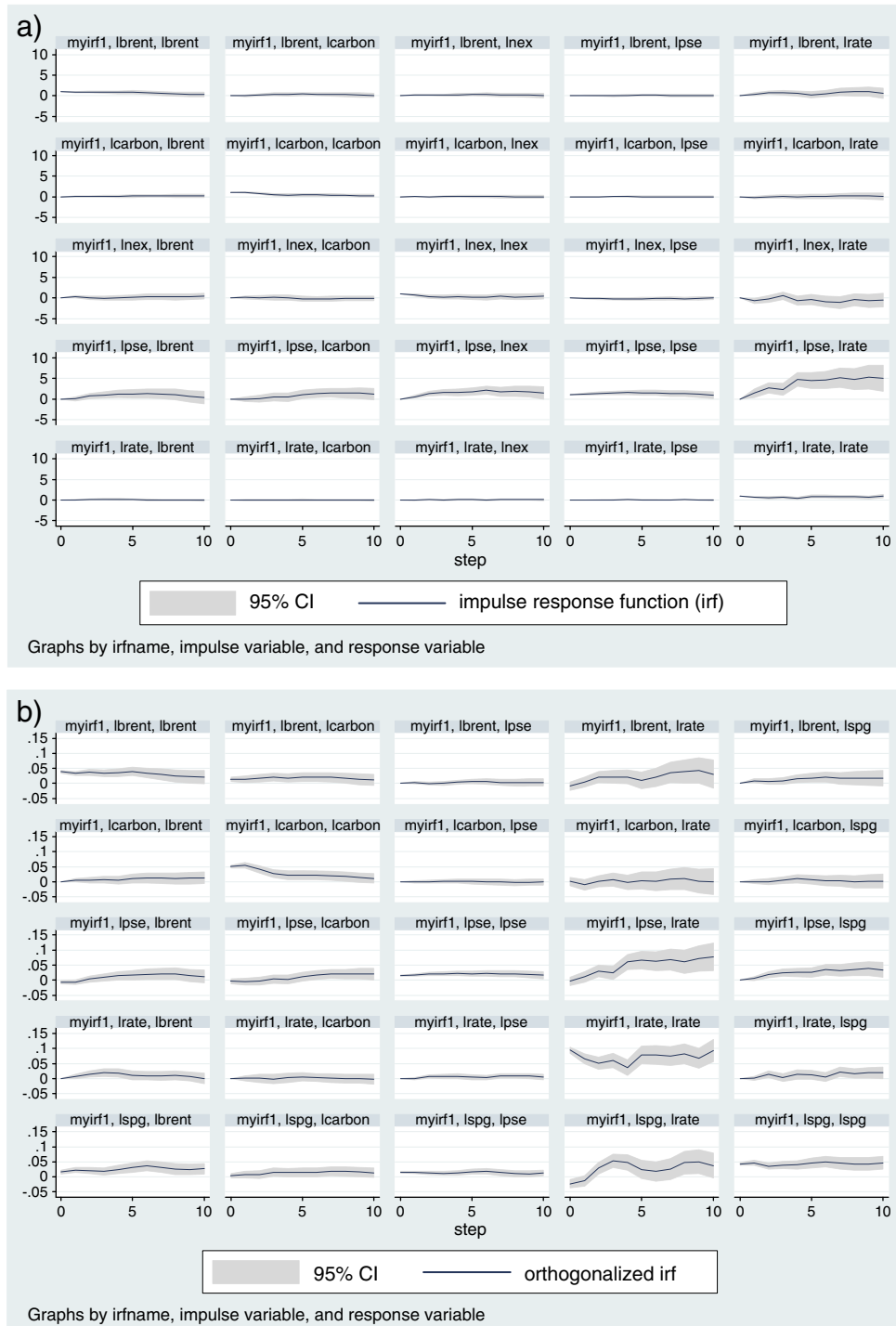


Fig. 4. a: impulse response functions (SPGCE). lnex: log of NEX, leco: log of ECO, lspg: log of SPGCE, lbrent: log of oil price, lcarbon: log of carbon price, lpse: log of PSE, and lrate: log of interest rate. b: orthogonal impulse response functions (SPGCE). lnex: log of NEX, leco: log of ECO, lspg: log of SPGCE, lbrent: log of oil price, lcarbon: log of carbon price, lpse: log of PSE, and lrate: log of interest rate.

response for the clean energy stocks caused by shocks to general technology stock prices. As expected, the impact of general technology stock prices on clean energy stocks is positive. This effect is immediate and continues for up to 10 weeks into the future. It is slightly stronger when measured by using clean stock prices derived from the SPGCE index relative to the NEX and ECO indices. The estimates of impulse response functions support the hypothesis that clean energy stocks and the stocks of high technology firms are viewed by the investors in parallel. The stock prices of clean energy

firms react positively to a shock to the prices of oil-producing firms. This positive effect is seen in week two, and it is positive and statistically significant. Note that it also remains positive in future weeks. However, it is statistically significant only in weeks two and four.

A shock to the interest rate variable causes a positive and significant impact on clean energy stock prices for a period of 2 weeks. After 2 weeks, the impact of the interest rate shock diminishes, and it does not show a particular trend thereafter.

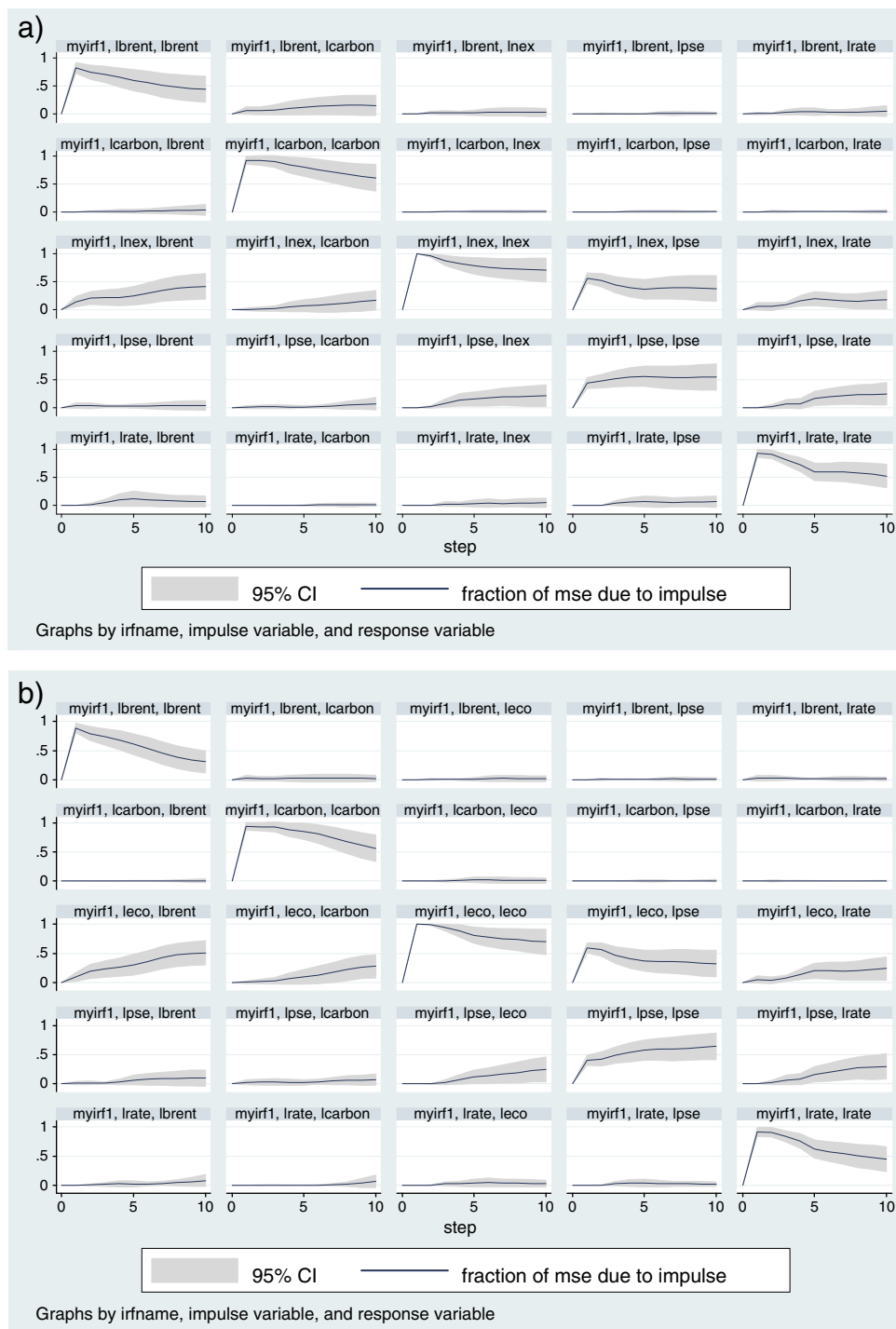


Fig. 5. a: variance decomposition analysis (NEX). lnex: log of NEX, leco: log of ECO, lspg: log of SPGCE, lbrent: log of oil price, lcarbon: log of carbon price, lpse: log of PSE, and lrate: log of interest rate. b: variance decomposition analysis (ECO). lnex: log of NEX, leco: log of ECO, lspg: log of SPGCE, lbrent: log of oil price, lcarbon: log of carbon price, lpse: log of PSE, and lrate: log of interest rate. c: variance decomposition analysis (SPGCE). lnex: log of NEX, leco: log of ECO, lspg: log of SPGCE, lbrent: log of oil price, lcarbon: log of carbon price, lpse: log of PSE, and lrate: log of interest rate.

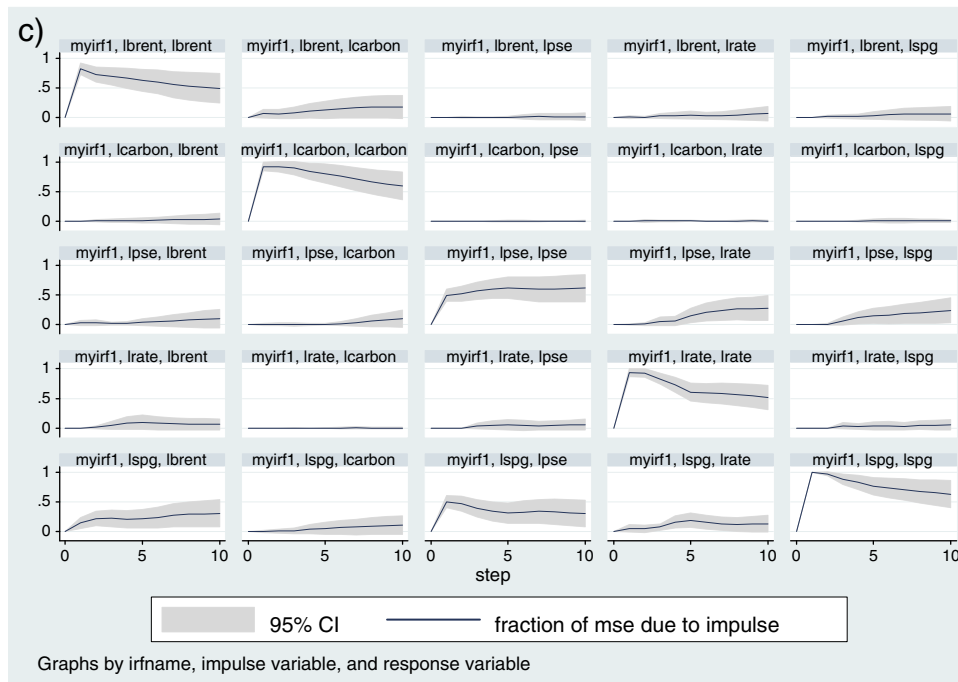


Fig. 5 (continued).

Because the Granger causality results show that carbon prices do not Granger cause movements in the prices of clean energy stocks, the impulse response function results reconfirm this relationship. Note that we observe quantitatively similar results for the shocks of all of the variables on the stock prices of clean energy firms, regardless of whether they are measured by using the NEX, ECO or SPGCE. Note also that we do not observe any systematic and statistically significant impact of the shocks in the prices of clean energy firms on the stock prices of high technology firms, oil prices, carbon prices, and interest rate variables.

For the sake of robustness, we also estimate the orthogonal impulse response functions. Qualitatively, the estimates of the orthogonal impulse response function are no different than the results of generalized impulse response functions. We observe that the impact of shocks to high technology stock prices is greater than the impact of shocks to oil prices.

Fig. 5 presents the results of the forecast error variance decomposition for all three models used in this study. The forecast error variance decomposition tells us the proportion of the movements in a sequence due to its own shocks versus shocks to the other variable. The variance decompositions suggest that shocks to the stock prices of general technology firms, oil price shocks and shocks to the rate of interest are considerable sources of volatility for many of the variables in the model. For clean energy stock prices, volatility in oil prices, general technology stock prices, and interest rates are the largest source of shock preceded by the variable itself. The contribution of these three variables to clean energy stock price variability is about 30% of the total variability.

5. Concluding remarks

This study analyzes the relationship between oil prices and alternate energy prices. We contribute to the literature by updating recent data, including those concerning the rise in oil prices and severe fall periods, and further considering the carbon market. This study finds that the variation in all three indices of clean energy stocks is explained by past movements in oil prices, stock prices of high technology firms and interest rates. We especially note that there is a

positive relationship because of rising oil prices and the substitution of alternate energy sources. In addition, we find that investors view the stocks of clean energy companies as they do the stocks of high technology firms.

Finally, carbon price returns are not a significant factor in stock price movements for clean energy firms. This result might be because carbon prices have been lower than oil prices. Therefore, carbon prices have not been able to create a stimulus for the switch to clean, low-carbon technologies from conventional fossil fuels.

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